WEEK 1: Definitions of Research

Research is generally referred to as an inquiry into a phenomenon. Research can also be defined as a systematic investigation to describe,explain, predict and control a phenomenon which employs inductive and deductive methods. Research is a systematic search for new knowledge and solutions to life issues. The Cambridge Dictionaries Online defines research as a “detailed study of a subject to discover information or achieve a new understanding of it.” Wikipedia defines research as the "creative and systematic work undertaken to increase the stock of knowledge. It involves collecting, organising, and analysing information to increase understanding of a topic or issue.

A research project may be an expansion of past work in the field. To test the validity of instruments, procedures, or experiments, research may replicate elements of prior projects or the project as a whole. It is opined that research in the library and information science is conducted to “solve professional problems, develop tools, and methods for the analysis of an organisation, services, and behaviour, to determine costs and benefits of our services, and most importantly, to establish or develop a body of theory on which to base our practice.”

The Merriam-Webster Online Dictionary, on the other hand, defines research as "studious inquiry or examination; especially investigation or experimentation aimed at the discovery and interpretation of facts, revision of accepted theories or laws in the light of new facts, or practical application of such new or revised theories or laws.” The Organisation for Economic Co-operation and Development (OECD) (2007) submits that research is “any systematic creative activity undertaken to increase the stock of knowledge, including knowledge of man, culture and society, and the use of this knowledge to devise new applications.”

Creswell (2008) defines research as “a process of steps used to collect and analyse information to increase our understanding of a topic or issue.” What can you deduce from all these definitions? The world cannot exist without research. Societies, to a very large extent, rise or fall based on the value they place on research. The process of research is systematic, logical, and organised. It involves taking conscious steps to carry out investigations that lead to relevant conclusions. It is also organised, which requires structure and methods that are repeatable or replicable. Research seeks answers to intelligent questions. Research studies help to arrive at valid and reliable information. This search for valid and reliable information is referred to as research. Organisations, institutions, and corporate entities depend on research to function efficiently and effectively.

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The research process leads to the generation of a body of evidence. From this, you can begin to appreciate the great value of research in your personal, career or professional life. From all these definitions and explanations,we deduce that research is a means of acquiring knowledge. The question then is, are there other means of acquiring knowledge? Our discussions below answer this question.

Methods of Acquiring Knowledge

Pause for a moment! Have you at any time considered how you acquire or acquired knowledge? How did you arrive at brushing your teeth every morning and night before bed? How did you arrive at the practice of making your bed in the morning? Think of how you could determine your friend is lying to you because of his strange behaviour and inability to look you in the eye. How did you choose and trust the sources of knowledge you have depended on all your life? Over the years, there have been different conventional and unconventional methods of acquiring knowledge. Men of old had their ways of acquiring knowledge that worked for them. Some of the methods of acquiring acknowledge include the following:

1. **Traditions**

Traditions is another potent source of knowledge acquisition. Society has relied on traditions in search of solutions to their problems and accept the verdict of tradition in the face of challenges. Societies that depended on tradition also accepted the culture and practice of their forefathers in the face of dealing with issues that affected their lives and survival. Knowledge acquired through traditions is not subjected to objectivity and tests. They are practices passed from one generation to another and sustained as the accepted means of acquiring knowledge. These days, these traditions are being eroded away in the face of economic and technological advancements. Interestingly, people who depended on traditions as a veritable source of knowledge begin to query and reject the practices once accepted as the final authority. Where do you come? What has been the role of tradition in knowledge acquisition in your locality? What has changed now?

2. Authority

Authority is the oldest and most familiar method of acquiring knowledge. From generation to generation, humanity has appealed to authority in knowledge acquisition. Society has been accepting new ideas as accurate and reliable because authority figures have declared them so.

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This is resorting to authorities in various specialities or fields of knowledge. The medical doctor is relied upon as the authority in medicine, and his verdict is accepted as the final authority. For instance, his advice to abstain from smoking because it is dangerous to health has become established knowledge. The dietician prescribes healthy meals to help you keep fit, and it is accepted as the final authority.The public, therefore, gains knowledge through the specialisation of the supposed experts. Nurses,economists, teachers, parents, priests and other religious leaders, the media, professors, and the government are accepted as an authority and great sources of knowledge. Authorities should be respected, but ideas should not always be accepted hook, line and sinker. Authorities often are wrong and misleading. You should learn to evaluate the credentials of authority figures. Many professors and medical doctors have misled people and led them to disastrous consequences.

3.Experience

Experience is another reliable and robust source of knowledge acquisition. Experience is a familiar and well-used source of acquiring knowledge. Society relies so much on experience. Our experiences and that of others are fundamental sources of knowledge from time immemorial. Generations have benefited from the resulting wisdom garnered from experience. Imagine where people or communities could not profit from the great wealth of experience; severe stagnation and retardation would have been. People talk of the wisdom of the ages, which generally amplifies that the ability to benefit from experience is one of the hallmarks of intelligent behaviour. We learn a lot from the occurrences and events that surround our existence. Remember the general saying that "experience is the best teacher." Is experience the best teacher? Consider this example: a laboratory technologist was running an experiment and had a chemical he derived from the mixture of several substances.

In his excitement, he decided to test the reaction of concentrated sulphuric acid when mixed with the new chemical. There was an immediate explosion. Fire gutted down the laboratory. The laboratory technologist was also killed. Is experience the best teacher? In this example, where is the student to explain the value of his experience? Experience is a good teacher. It would help if you accepted it and then rely on the certified experiences of others, especially those who are ahead of you in life, career, or profession. Remember, the main thing here is the experience you rely on must be certified. If you follow the foolish man's experience, you will end up in the stupid man's destination. Therefore, you should understand clearly that experience has limitations. Your personality characteristics, traits, perceptions, and general disposition to life affect the experience you acquire. Two persons in the same situation could

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obtain different experiences. The blind men and the elephant readily come to mind. Two supervisors observing the same classroom simultaneously could truthfully compile very different reports if one focused on and reported what went right. The other focused on and reported the things that went wrong." And all of them are records of experience.

4. **Trial and error**

Although this sub-area says trial and error, it is one of the fundamental methods of acquiring knowledge. It is a method characterized by multiple attempts until success is achieved. When you employ trial and error, it involves trying, observing, and repeating the process until success is reached. Trial and error have been professionally described as a means of "problem-solving, repair, tuning or obtaining knowledge" (Wikipedia). A means of acquiring knowledge is known as a brute force in which you generate and test or a means that enables you to guess and check. It is only efficient when the attempts are based on prior knowledge and information. In his study of the law of effect in learning, the first initiator of the trial-and-error theory was Edward Lee Thorndike. Interestingly, several professionals employ trial and error in handling technical issues, which are sometimes successful and at other times frustrating. Do you know that medical doctors sometimes eliminate causes and illnesses through trial and error? Take for instance, I went to the hospital and complained of weakness, temperature, and headache. The Doctor asked a few questions, touched my body, and prescribed certain drugs. I religiously followed the doctor's prescription, and after six days, I had a mild relief, but the symptoms were still disturbing me. How did the doctor arrive at his initial prescription? He assumed I must be suffering from malaria and recommended the prescription to take care of it.Seven days after, the same doctor discovered that it was not malaria that was affecting me. He has isolated malaria by trial and he was not correct. This time, he recommended a laboratory test.

5. **Intuition**

This is another method of knowing. Intuition relies on guts, emotions, and instincts. It does not rely on rational thought; it accepts what feels true. Have you had a course to rely on intuition, and how did it guide you in making decisions? Intuitions can be wrong because they are propelled by "cognitive and motivational biases rather than logical reasoning or scientific evidence". However, several people depend on intuition and accept it as superior to decisions based on analysis.

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6. **Reasoning**

This is another important source of acquiring knowledge. Many people arrive at decisions based on abstract thinking or associating relationships between events and variables. Reasoning enables you to collect facts, generalize and make out cause to effect or effect to cause, and derive conclusions from premises. It assembles data, analyses, findings and decides. It takes you to the door of final assessment and judgement.Reasoning is a powerful source of knowledge acquisition but can be very misleading. Your reasoning can be affected by circumstances and vicissitudes of life that may eventually cause your reasoning to be warped, corrupted and crooked. God has given you the power of reason but also allowed you to substantiate your reasoning with other available credible sources to prevent you from going into error. Several persons have dangerously erred based on a complete reliance on their limited reasoning.

7. **Research:** The scientific method

The most reliable and valid method of acquiring knowledge is the scientific method, also referred to as research. This involves the systematic and logical process of collecting, analysing, and evaluating data to answer questions and test hypotheses. Here, the hypothesis is regarded as an intelligent research guess. Although the scientific method is the most authentic way of knowing, intuition, authority, and experience still play significant roles. However, the scientist further employs systematic steps under controlled conditions to make careful observations to arrive at valid conclusions. The scientific method is characterised by objectivity, replicability, validity, reliability, purposiveness, precision, and accuracy. The scientific method is accurate and reliable but tedious, time-consuming, and costly. It primarily addresses empirical questions and cannot answer immediate and instantaneous questions.

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